

```
import requests
import geopandas as gpd
import pandas as pd
```

```
###Open County data
```

```
from osgeo import gdal
gdal.SetConfigOption('SHAPE_RESTORE_SHX', 'YES')

# https://catalog.data.gov/dataset/2024-cartographic-boundary-file-shp-county-subdivision-for-united-states-1-500000
# https://stackoverflow.com/questions/61436956/set-shape-restore-shx-config-option-to-yes-to-restore-or-create-it
# requires both .shp and .shx files in file directory
file_path = "/content/cb_2024_us_county_500k.shp"
counties_gdf = gpd.read_file(file_path)
```

```
### Get Ethnic Data
```

```
import json

#r = requests.get("https://api.census.gov/data/2023/acs/acs5/groups/B02018.json")
# uploading saved B02018.json file since Census API isn't working on 2026.01.14
with open('B02018.json', 'r') as f:
    columns_obj = json.load(f)
```

```
###Get columns to query and rename for later
columns = []
rename_vars = {}
variables = columns_obj["variables"]

for name, variable in list(variables.items()):
    v_split = variable["label"].split("!!")
    if len(v_split) < 3:
        continue

    if v_split[0] == "Estimate":
        label = v_split[-1]
        rename_vars[name] = label

    if (name.endswith("E") or name.endswith("M")) and v_split[-2] == "East Asian:":
        columns.append(name)

    if (name.endswith("E") or name.endswith("M")) and v_split[-2] == "Southeast Asian:":
        columns.append(name)

print(columns)
# https://www.geeksforgeeks.org/python/check-if-element-exists-in-list-in-python/
#print(any(x == "B02018_016E" for x in columns))
#print(any(y == "B02018_016M" for y in columns))
```

```
['B02018_016M', 'B02018_017E', 'B02018_017M', 'B02018_018E', 'B02018_014M', 'B02018_015E', 'B02018_015M', 'B02018_016E', 'B02018_018M',
```

```
df = pd.read_csv("formtted_df.csv")
```

```
print("df.columns")
print(df.columns)
```

```
df.columns
Index(['Unnamed: 0', 'GEOIDFQ', 'B02018_017E', 'B02018_018E', 'B02018_015E',
      'B02018_016E', 'B02018_019E', 'B02018_012E', 'B02018_013E',
      'B02018_014E', 'B02018_010E', 'B02018_011E', 'B02018_005E',
      'B02018_006E', 'B02018_003E', 'B02018_004E', 'B02018_009E',
      'B02018_007E', 'B02018_008E', 'B02018_002E', 'B02018_020E'],
      dtype='object')
```

```

moe_data_all = pd.read_csv("ACSDT5Y2023.B02018-2026-01-14T233106.csv")
#print(moe_data_all.shape)
# https://pandas.pydata.org/pandas-docs/stable/reference/api/pandas.DataFrame.transpose.html
moe_data_tr = moe_data_all.transpose()
#print(moe_data_tr.head())

# get rows of transposed dataframe
print(moe_data_tr.iloc[0])
print(moe_data_tr.columns)

# drop all columns corresponding to South Asian or Central Asian

moe_data_tr.drop(columns=["Unnamed: 0"], inplace=True)

```

```

0          Total Groups Tallied:
1          East Asian:
2      Chinese, except Taiwanese
3          Hmong
4          Japanese
5          Korean
6          Mongolian
7          Okinawan
8          Taiwanese
9      Other East Asian
10     Southeast Asian:
11         Burmese
12         Cambodian
13         Filipino
14         Indonesian
15         Laotian
16         Malaysian
17         Mien
18         Singaporean
19         Thai
20         Vietnamese
21     Other Southeast Asian
22         South Asian:
23         Asian Indian
24         Bangladeshi
25         Bhutanese
26         Nepalese
27         Pakistani
28         Sikh
29         Sri Lankan
30     Other South Asian
31         Central Asian:
32         Kazakh
33         Uzbek
34     Other Central Asian
35     Other Asian, specified
36     Other Asian, not specified
Name: Label (Grouping), dtype: object
RangeIndex(start=0, stop=37, step=1)

```

